CLARA-Care: A Governed Longitudinal Health-AI Prototype Linking Interoperable Evidence, Task-Bounded Context, and Persistent Writeback

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Abstract. CLARA-Care is a research prototype for longitudinal personal-health AI that connects source-preserving multimodal data ingestion, task-bounded model context, interactive verification, and governed persistent writeback. The live demonstration guides attendees through an end-to-end clinical workflow: (1) automated prescription digitisation via Gemini 3.7 Flash Multimodal OCR into FHIR-standardised longitudinal timeline events; (2) task-bounded clinical context assembly (THSS) with an interactive **Claim & Evidence Inspector** exposing zero-PHI record links and FIDES dual-layer safety verification; and (3) multi-agent state writeback with **Dynamic Wound-Wait DAG Locking** preventing deadlocks and lock thrashing during concurrent record updates. The system remains a research prototype and is not deployed for clinical care.

Purpose and problem. Longitudinal health assistants repeatedly ingest heterogeneous clinical records and propose durable updates to patient state. Relevant retrieval alone neither guarantees that the model sees fresh, task-scoped clinical context nor prevents stale or unauthorized writeback if clinical state, consent, or policy drifts during inference. Furthermore, multi-agent workflows touching shared health profiles risk concurrency bottlenecks and deadlocks under monolithic locking. CLARA-Care addresses these challenges by making context boundaries, provenance chains, and transactional locking explicit across the read-to-write lifecycle, preventing direct unmediated model writeback to canonical state.

System and live demonstration. The live demonstration showcases three core components operating on synthetic and benchmark patient records:

- *1. Multimodal OCR & Timeline Ingestion:* Attendees observe live prescription scanning powered by Gemini 3.7 Flash Tiered OCR. The pipeline extracts complex medication entities (names, strengths, frequencies, routes) from handwritten hospital prescriptions and packaging (0.981 drug F1 score), mapping them into bitemporal FHIR STU3/R4 records while separating clinical valid time (τ_{valid}) from transaction time (τ_{known}).
- *2. Real-Time Claim & Evidence Inspector:* During interactive clinical Q&A and decision support, the web interface provides a slide-out Claim & Evidence Inspector. It reveals the complete reasoning chain, attributing each generated clinical claim to underlying zero-PHI personal health observations and authoritative medical guidelines (Vietnam National Pharmacopoeia, MOH guidelines) verified via the FIDES safety engine with 99.4% empirical confidence.
- *3. Dynamic Wound-Wait DAG Locking in Action:* When concurrent agent proposals execute multi-hop entity expansions, the system enforces Kung-Robinson OCC over lexicographically ordered entity partition keys ($\prec_{\text{lex}}$) and the Rosenkrantz et al. Wound-Wait non-preemptive priority protocol ($ts(T_i)$). Older transactions preemptively wound younger lease holders, eliminating deadlock cycles (0.00% deadlock rate) while avoiding monolithic profile lock thrashing.

The demo walks through three live scenarios: a clean valid commit, a stale-state abort, and an active consent-revocation rejection.

Evaluation evidence. In a frozen 12-schedule PostgreSQL governance-TOCTOU matrix, the prototype observed no forbidden commit across consent revocation, role change, policy-epoch advance, state change, and compound drift. In formal verification, bounded exploration checked 21,361 states and 90,432 transitions to depth 5 with 0 violations of 11 specified safety invariants. In CareGuard-VN multimodal evaluation across $N = 1{,}500$ Vietnamese clinical test cases, Gemini 3.7 Flash OCR achieved 0.981 drug F1 and 99.6% severe DDI sensitivity with 100% fail-closed FIDES safety blocking. In dynamic concurrency stress benchmarks, the Wound-Wait DAG locking engine scaled throughput without thrashing, maintaining a 0.00% deadlock rate across high-contention multi-agent workloads. In a 64-subject synthetic cohort, Strict THSS achieved all-axis exact match for 63/64 Claude and 64/64 Gemini calls. These are synthetic and software-level evaluations, not clinical validation.

Innovation and current deployment status. The system unites FHIR-interoperable ingestion, multimodal prescription OCR, task-bounded context projection, transparent claim inspection, and deadlock-free Wound-Wait transactional writeback into a single governed architecture. CLARA-Care is an open student-led research prototype under active isolated testing; it has not been deployed for patient care, and makes no claim of clinical efficacy or regulatory compliance.

References. [1] HL7 International, FHIR R4, 2019. [2] W3C, PROV-O, 2013. [3] D. J. Rosenkrantz, R. E. Stearns, P. M. Lewis, ACM TODS, 3(2):178–215, 1978. [4] H. T. Kung, J. T. Robinson, ACM TODS, 6(2):213–226, 1981. [5] X. Li et al., "MemTX," arXiv:2607.23929, 2026. [6] I. Santos-Grueiro, "Commit-Time Authorization," arXiv:2607.10487, 2026.